EDUHUB / ENGINEERING DOCUMENTATION

ACTIVITY DIAGRAMS

EduHub | Online Learning Platform

16	03	48
UML WORKFLOWS	CORE USER ROLES	USE-CASE BASELINE

Workflow logic, decision guards, alternate paths, parallel actions, permissions and persistence effects.

Based on the supplied EduHub brief and previously developed architecture, use cases, requirements, DFD and sequence diagrams.

SYSTEM ANALYSIS & DESIGN / NOVEMBER 2026

Behavior model | executive overview

A workflow-oriented companion to the EduHub use-case and sequence models.

16	08	03	01
ACTIVITY VIEWS	DATA DOMAINS	APPLICATION ROLES	SOURCE BRIEF

Purpose and modeling boundary

This document describes what actions occur and under which guards for the key EduHub journeys. The activity views track control flow and business decisions, unlike the DFD (information movement) or the sequence diagrams (participant message ordering).

Model scope

INCLUDED

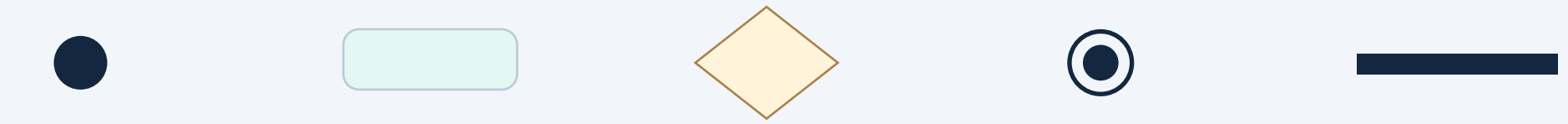
16 operational activities; branching, loops, exit conditions, a concurrent analytics path, role ownership, proposed controls, and scenario mappings.

NOT INCLUDED

SQL DDL, detailed REST endpoint contracts, state invariants for every entity, UI mockups, implementation claims, or cloud configuration scripts.

UML activity notation & decisions

UML 2.5.1 activity semantics, practical diagram legend, and intentional design assumptions. [R1]



Initial | Action | Decision | Final | Fork / join

How to interpret the flows

Rounded actions	A unit of work performed by the labeled role or component.
Guarded decision	Exactly one outgoing path is taken when its condition is true.
Loop / retry	Re-entry is permitted only after re-evaluating the indicated guard.
Fork and join	Multiple logical branches execute before the joined path continues.
Activity final	Ends the modeled workflow; not necessarily the user session.

Design status: process outcomes derive from the EduHub brief; signature checking, idempotency, retries, authorization specifics and role policies are proposed engineering controls, not implemented features.

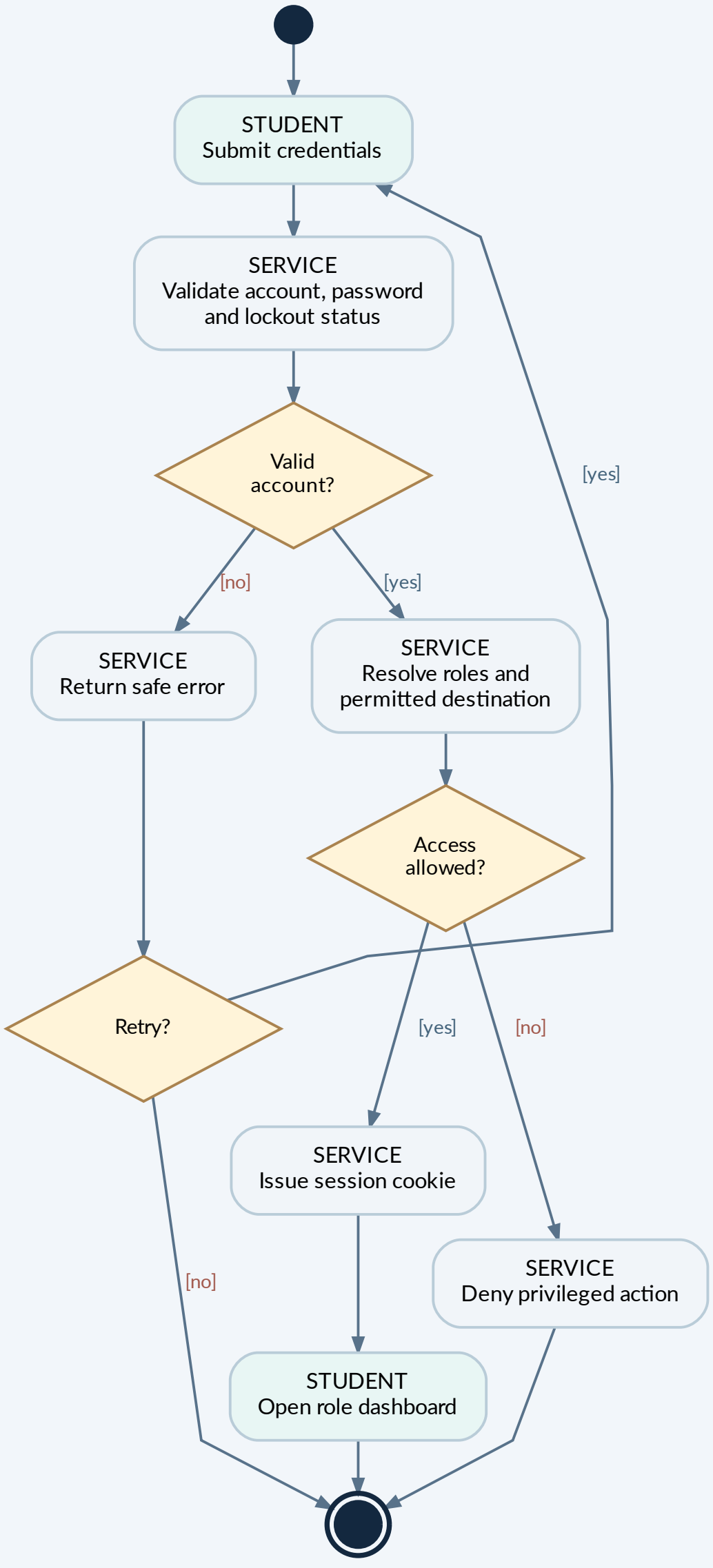
Activity catalog & domain ownership

Each view has a stable ID aligned with the earlier EduHub sequence-model numbering.

AD-01	Account sign-in & authorization	IDENTITY
AD-02	Create or edit a course draft	COURSE AUTHORING
AD-03	Upload and process lesson video	MEDIA PIPELINE
AD-04	Submit, review and publish a course	MODERATION
AD-05	Browse catalog and enroll for free	DISCOVERY & ENROLLMENT
AD-06	Cart, discount and Stripe checkout	COMMERCE
AD-07	Verify Stripe webhook and fulfill order	PAYMENT FULFILLMENT
AD-08	Authorize and stream protected lesson	SECURE LEARNING
AD-09	Record lesson completion and progress	LEARNING PROGRESS
AD-10	Take, grade and complete a quiz	ASSESSMENT
AD-11	Generate completion certificate	CERTIFICATION
AD-12	Public certificate verification and sharing	CERTIFICATE VERIFICATION
AD-13	Course Q&A and learner reviews	ENGAGEMENT
AD-14	Build dashboard analytics and reports	REPORTING
AD-15	Manage lesson notes and bookmarks	PERSONAL LEARNING
AD-16	Admin manage users, taxonomy and settings	PLATFORM GOVERNANCE

Account sign-in & authorization

AD-01 / IDENTITY / UC-02 / Brief pp. 1-2



Student **Service**

NOMINAL OUTCOME

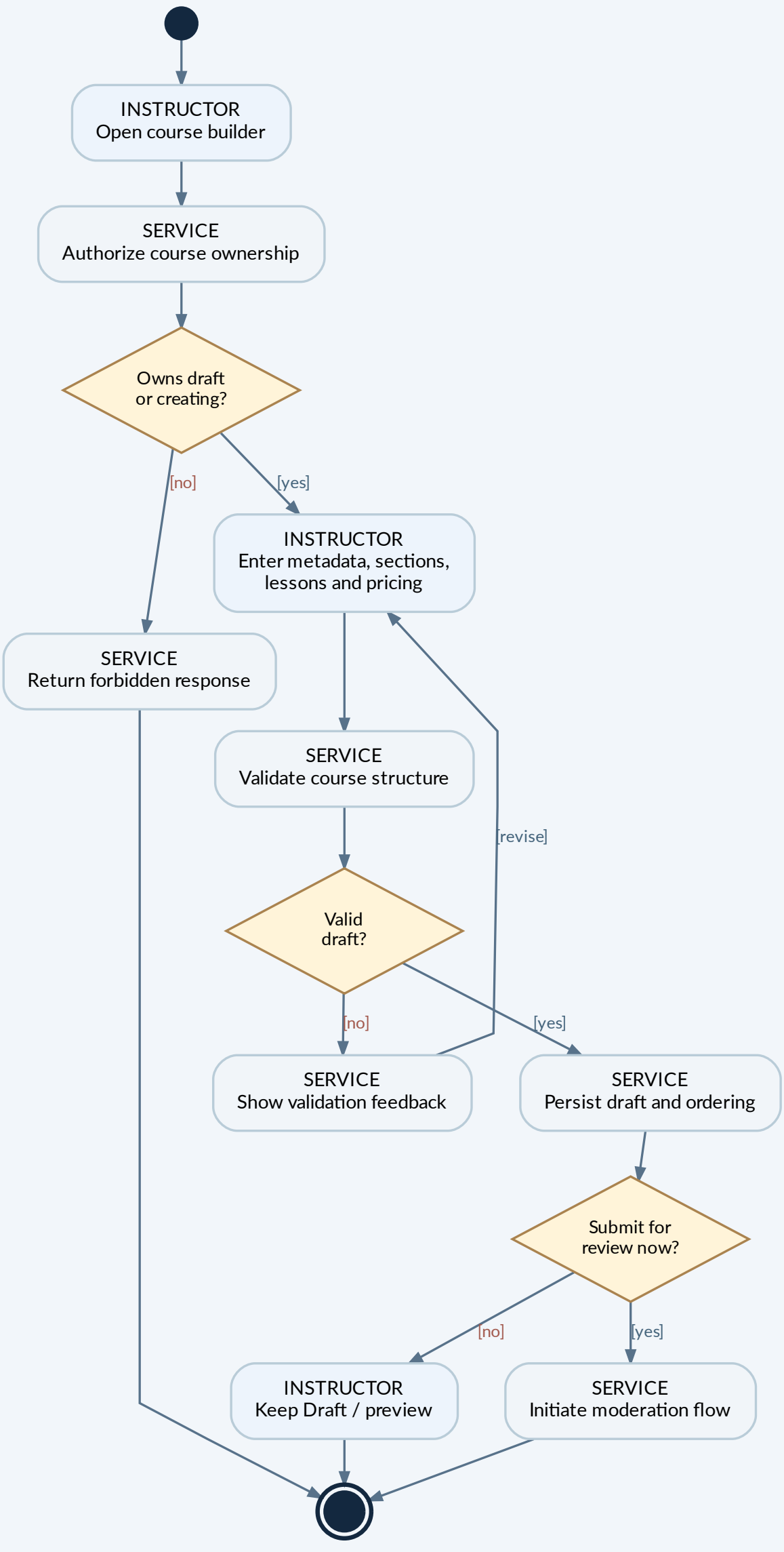
Authenticated user reaches an authorized role-specific entry point.

ALTERNATE / EXCEPTION

Invalid / locked account shows generic denial; unauthorized resource access is blocked.

Create or edit a course draft

AD-02 / COURSE AUTHORING / UC-25, UC-26, UC-29, UC-30 / Brief pp. 1, 3



Instructor

Service

NOMINAL OUTCOME

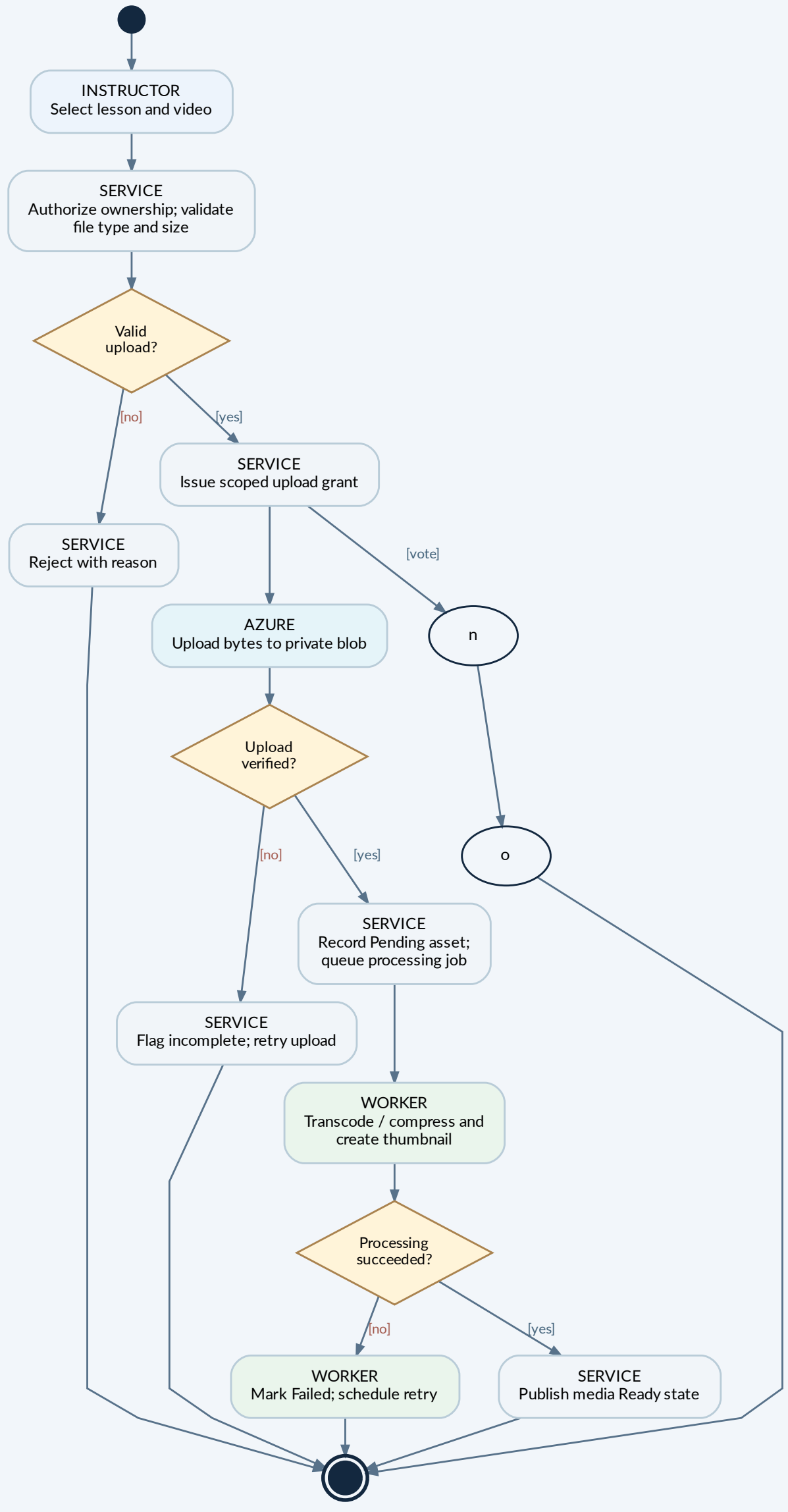
Course sections, lesson order, pricing and metadata persist as a draft.

ALTERNATE / EXCEPTION

Ownership violation denies mutation; invalid metadata returns actionable errors.

Upload and process lesson video

AD-03 / MEDIA PIPELINE / UC-27, UC-28, UC-46 / Brief p. 3



Instructor | Service | Azure | Worker

NOMINAL OUTCOME

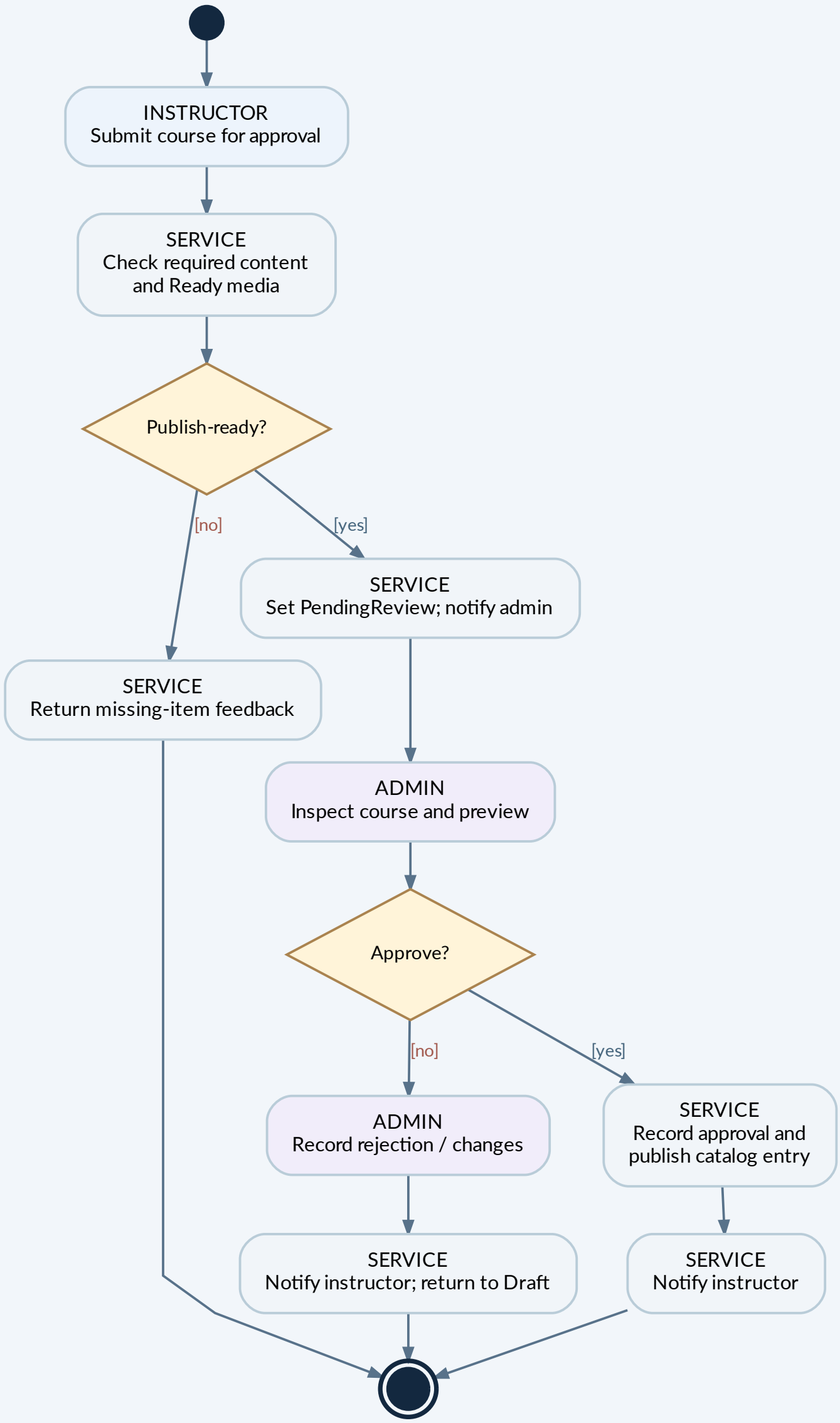
Verified, processed video is Ready and addressable via private media metadata.

ALTERNATE / EXCEPTION

Invalid file, interrupted upload or processor failure leaves no playable Ready asset.

Submit, review and publish a course

AD-04 / MODERATION / UC-31, UC-37, UC-38, UC-48 / Brief pp. 3, 5



Instructor | Service | Admin

NOMINAL OUTCOME

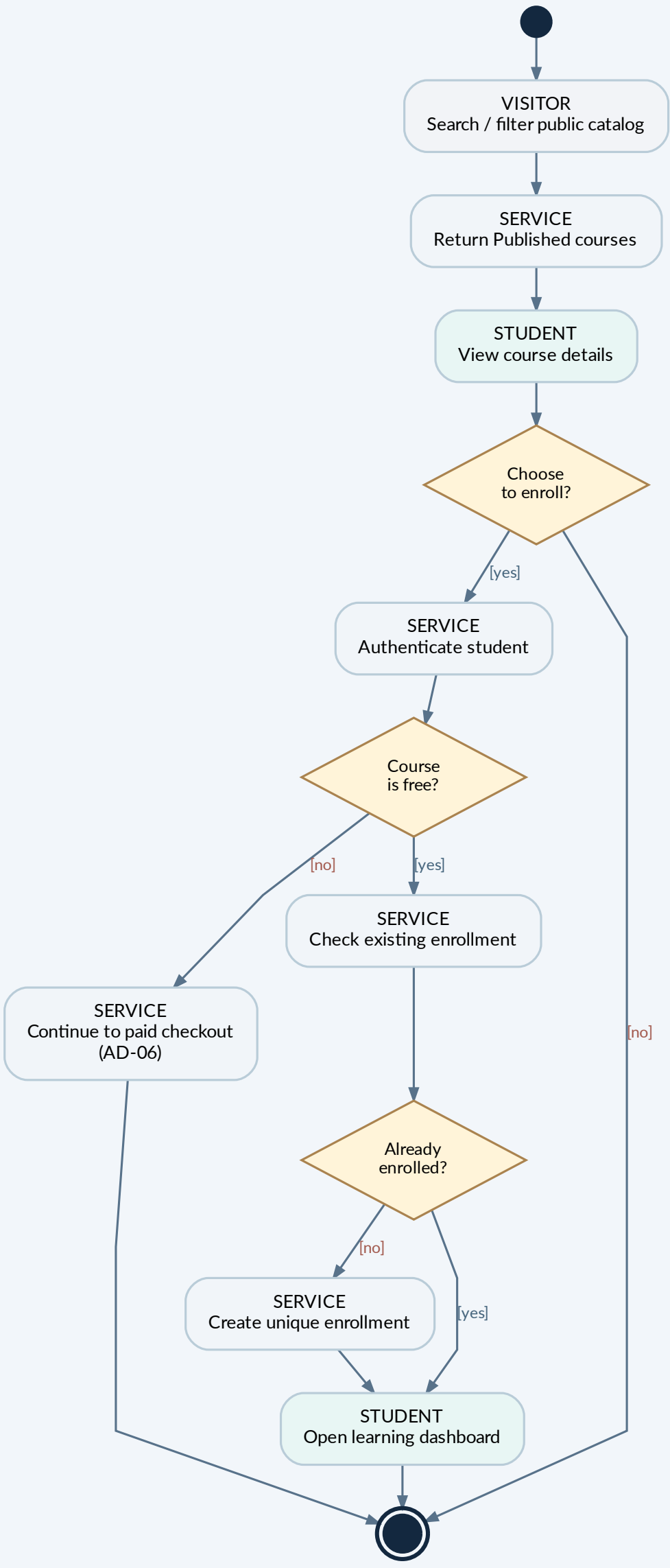
Admin-approved course becomes visible in the public catalog.

ALTERNATE / EXCEPTION

Incomplete submission is blocked; rejected course remains unavailable for sale.

Browse catalog and enroll for free

AD-05 / DISCOVERY & ENROLLMENT / UC-03, UC-04, UC-07 / Brief pp. 1, 4



Visitor **Service** **Student**

NOMINAL OUTCOME

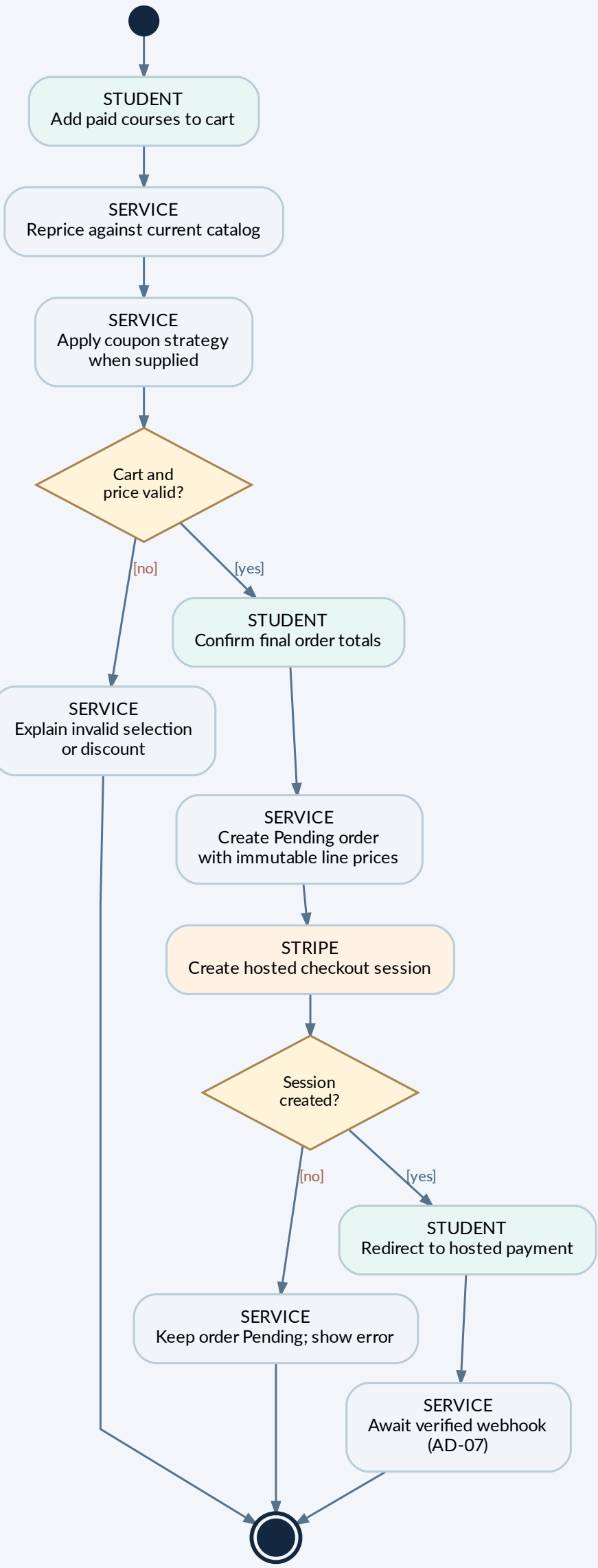
Free course receives one enrollment, without a payment requirement.

ALTERNATE / EXCEPTION

Paid course routes to checkout; repeat enrollments do not duplicate records.

Cart, discount and Stripe checkout

AD-06 / COMMERCE / UC-08, UC-09, UC-10 / Brief p. 4



Student **Service** **Stripe**

NOMINAL OUTCOME

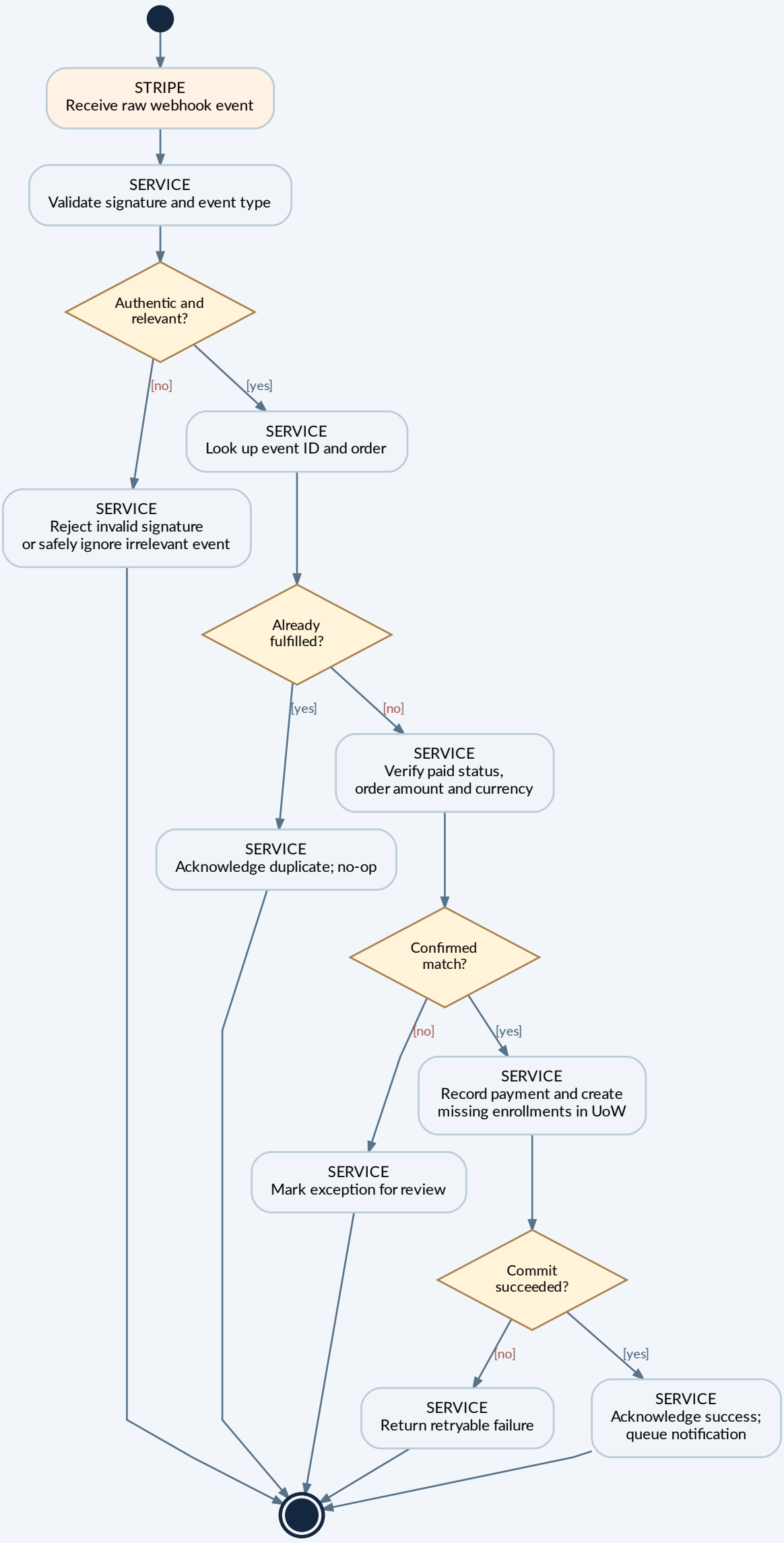
Pending order and Stripe checkout session exist; payment is not yet fulfilled.

ALTERNATE / EXCEPTION

Invalid coupon, stale prices, session error or cancellation does not create paid access.

Verify Stripe webhook and fulfill order

AD-07 / PAYMENT FULFILLMENT / UC-43, UC-44, UC-48 / Brief p. 4



Stripe

Service

NOMINAL OUTCOME

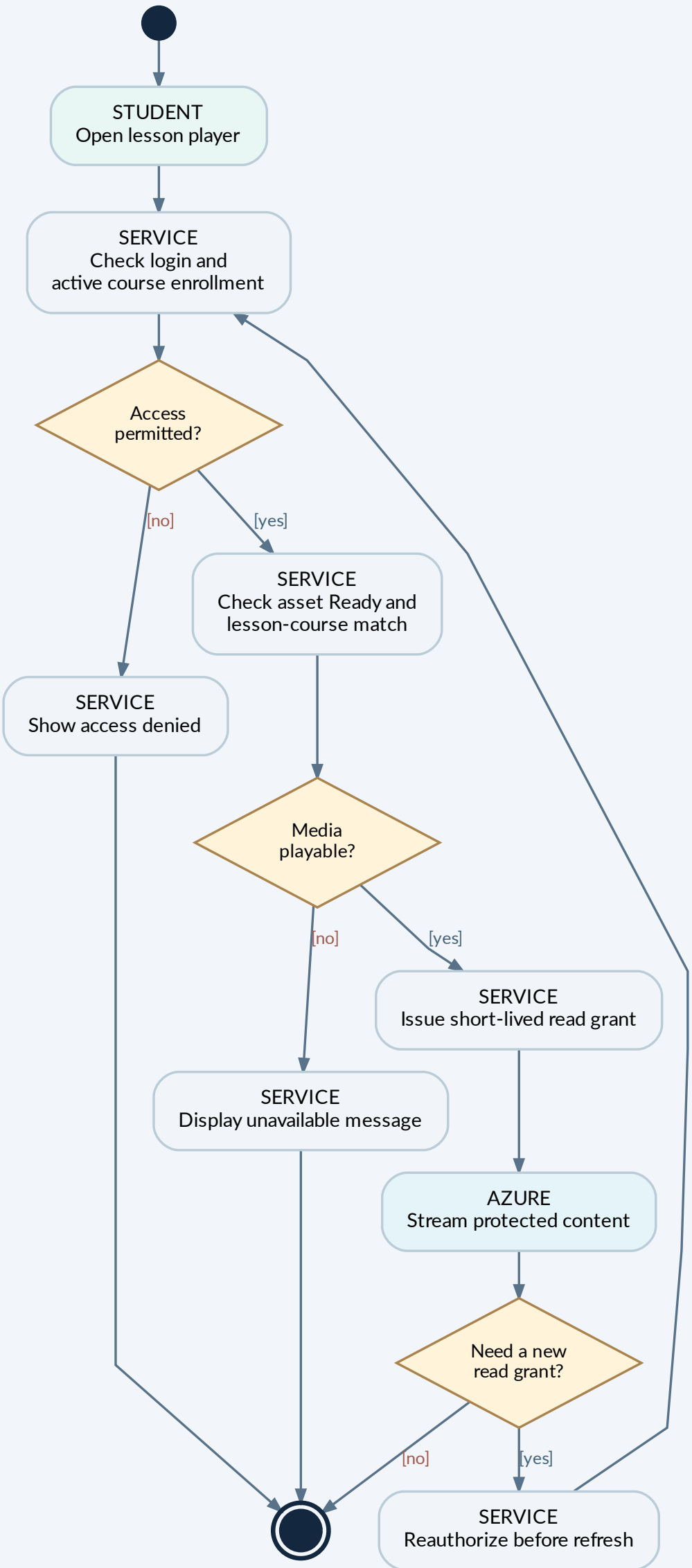
Verified payment grants entitlements once and commits atomically.

ALTERNATE / EXCEPTION

Forged, duplicated, unmatched or failed events do not create duplicate enrollments.

Authorize and stream protected lesson

AD-08 / SECURE LEARNING / UC-12, UC-13, UC-47 / Brief pp. 3-4



Student | Service | Azure

NOMINAL OUTCOME

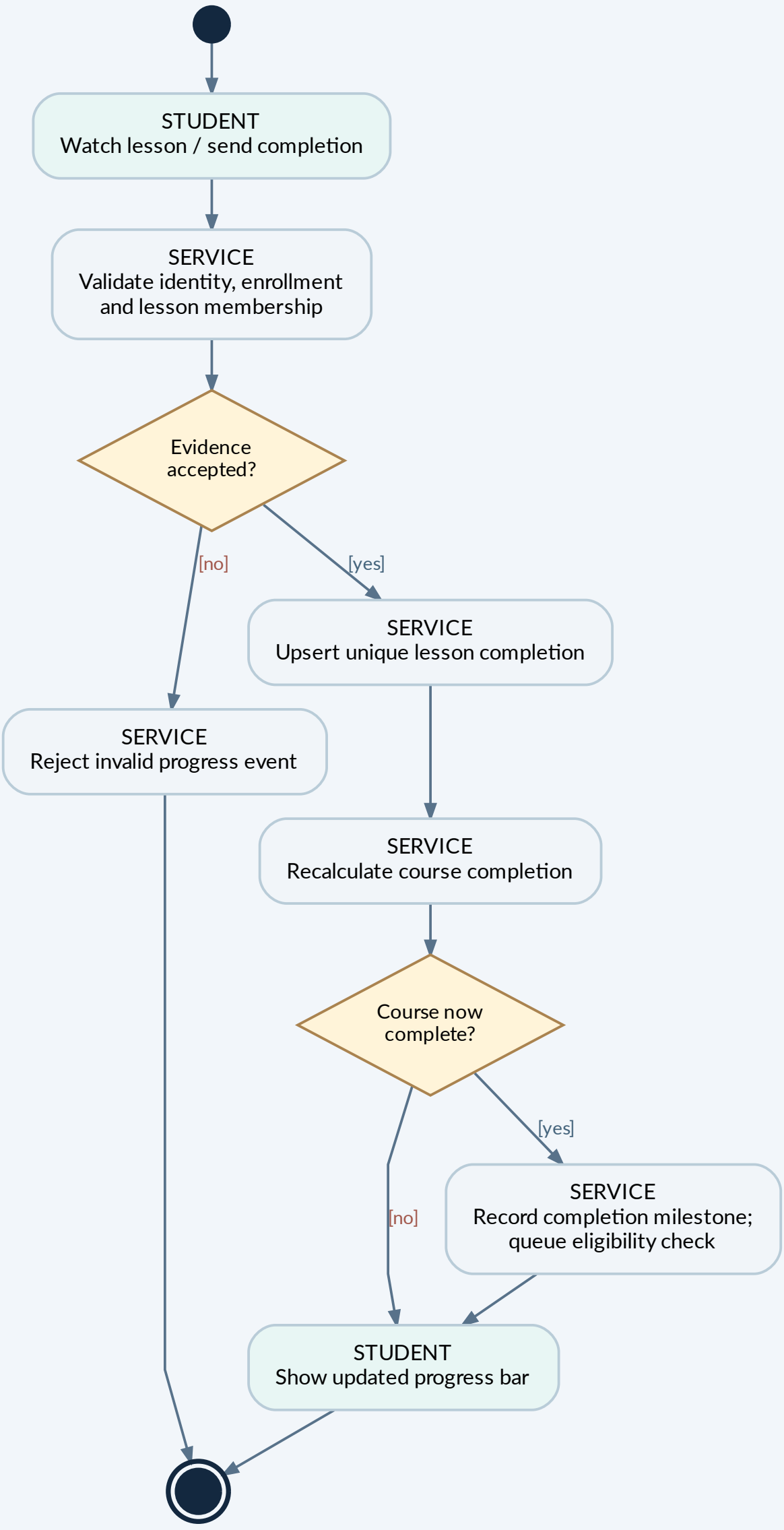
Authorized learner receives short-lived access to playable media.

ALTERNATE / EXCEPTION

Unenrolled user or unready media receives no usable grant.

Record lesson completion and progress

AD-09 / LEARNING PROGRESS / UC-14, UC-15, UC-48 / Brief p. 4



Student **Service**

NOMINAL OUTCOME

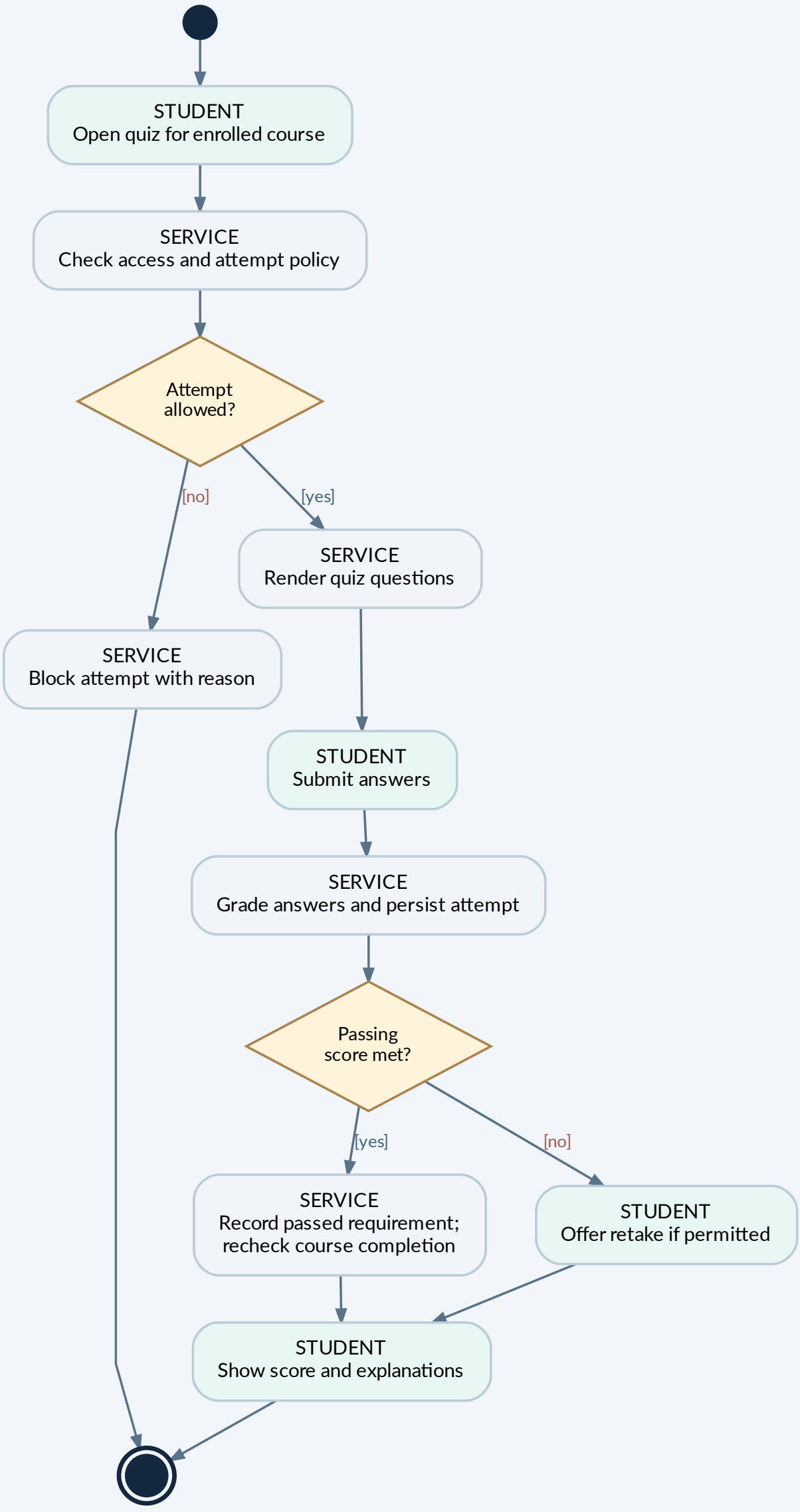
Progress updates without duplicate counting; completion may trigger certification check.

ALTERNATE / EXCEPTION

Stale or unauthorized events are rejected and repeated completions are safe.

Take, grade and complete a quiz

AD-10 / ASSESSMENT / UC-16, UC-33 / Brief p. 5



Student **Service**

NOMINAL OUTCOME

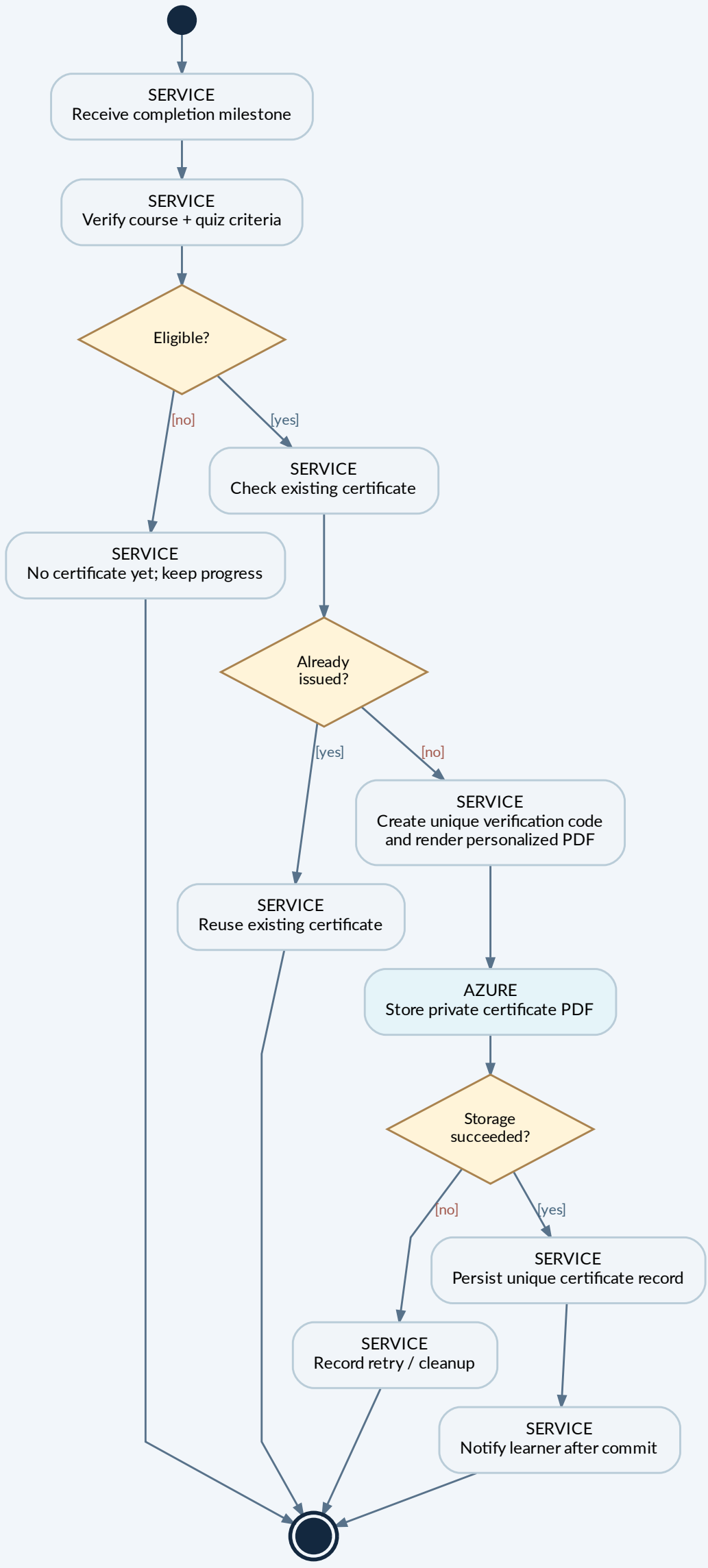
Attempt stored, score displayed, pass status contributes to eligibility.

ALTERNATE / EXCEPTION

Unauthorized or exhausted attempt blocked; failed quiz offers retake only if policy allows.

Generate completion certificate

AD-11 / CERTIFICATION / UC-23, UC-45 / Brief p. 5



Service Azure

NOMINAL OUTCOME

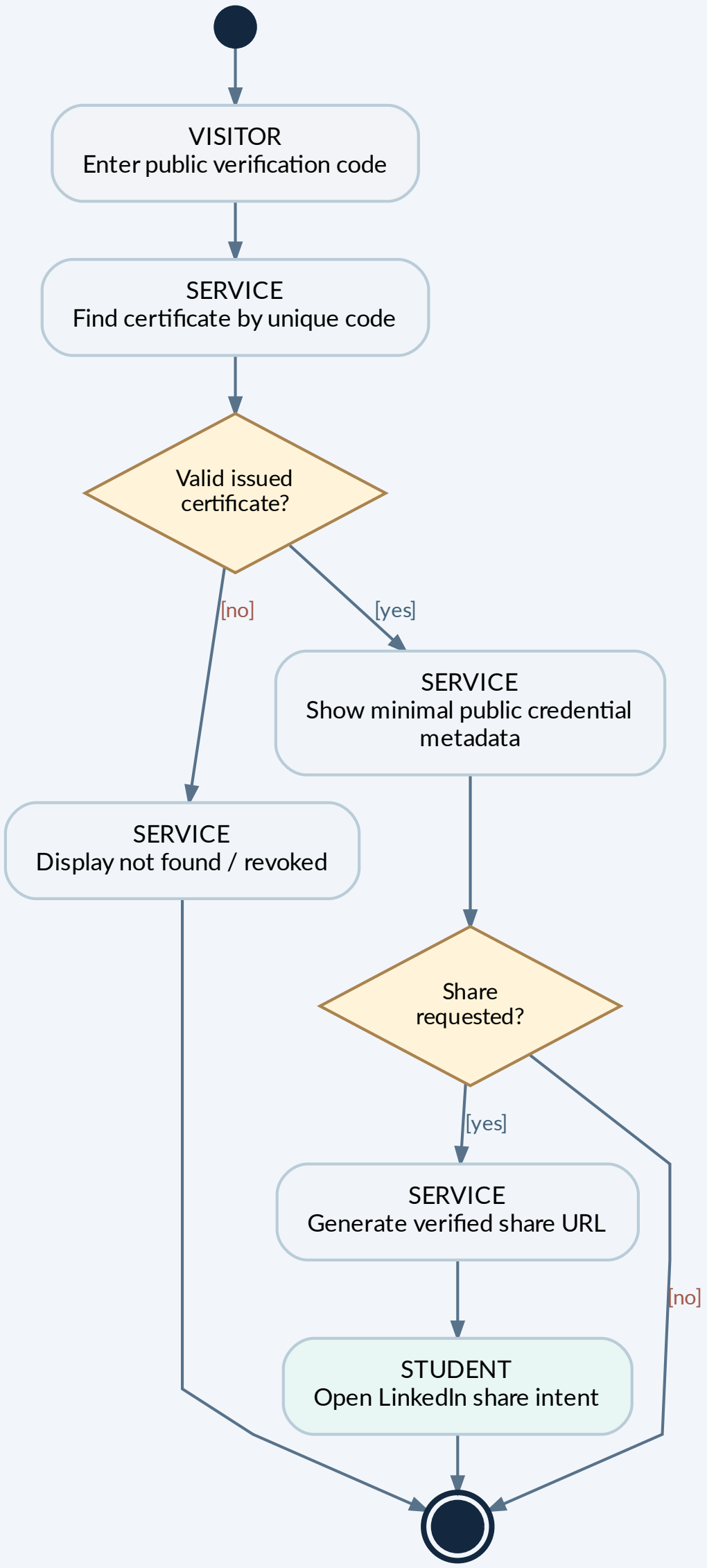
One verifiable personalized PDF certificate is registered for the learner.

ALTERNATE / EXCEPTION

Ineligible, duplicate or failed-storage case issues nothing new.

Public certificate verification and sharing

AD-12 / CERTIFICATE VERIFICATION / UC-05, UC-23, UC-24 / Brief p. 5



Visitor **Service** **Student**

NOMINAL OUTCOME

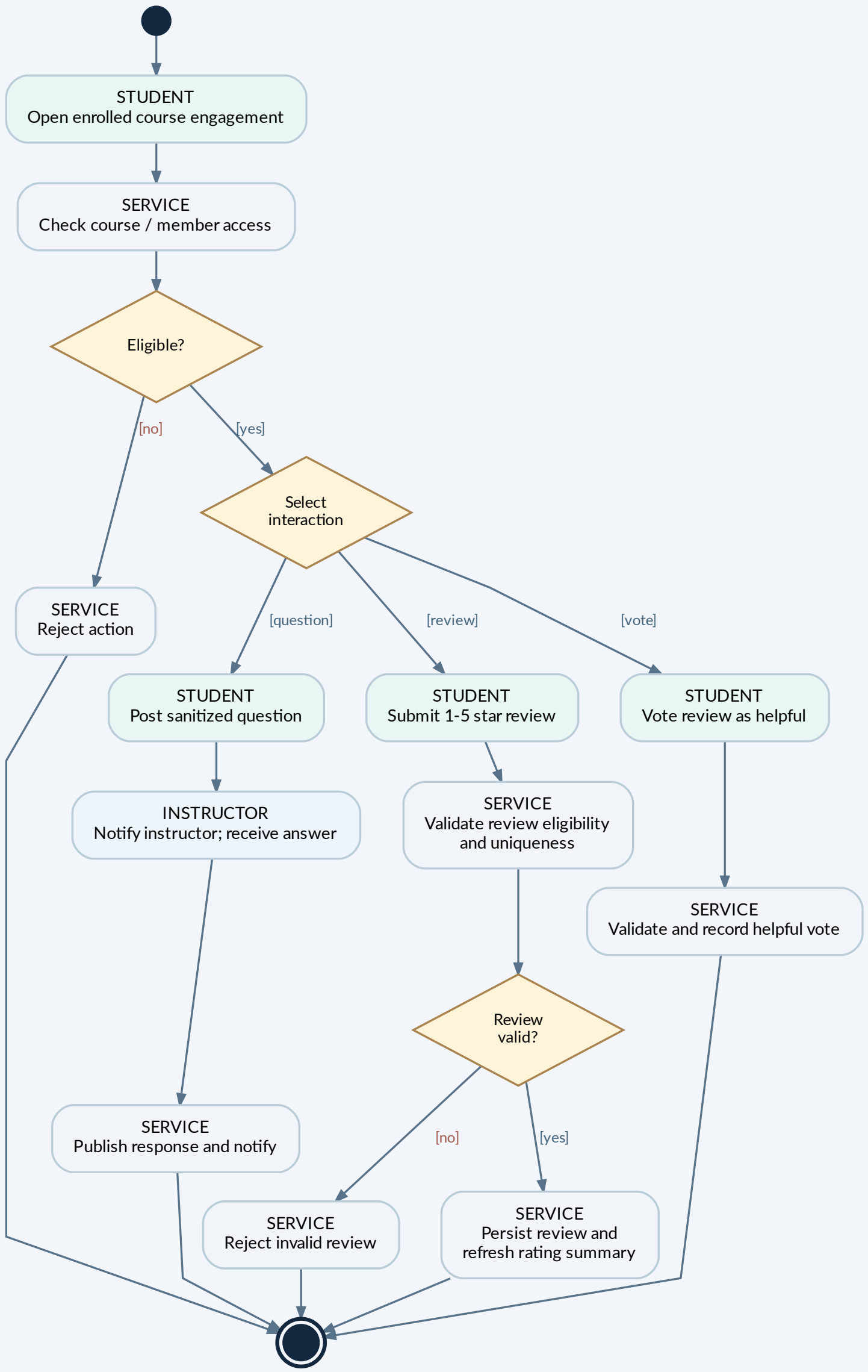
Valid credential is publicly verifiable; learner can initiate LinkedIn sharing.

ALTERNATE / EXCEPTION

Unknown or invalid code returns safe generic feedback.

Course Q&A and learner reviews

AD-13 / ENGAGEMENT / UC-19, UC-21, UC-22, UC-35 / Brief pp. 3, 5



Student | Service | Instructor

NOMINAL OUTCOME

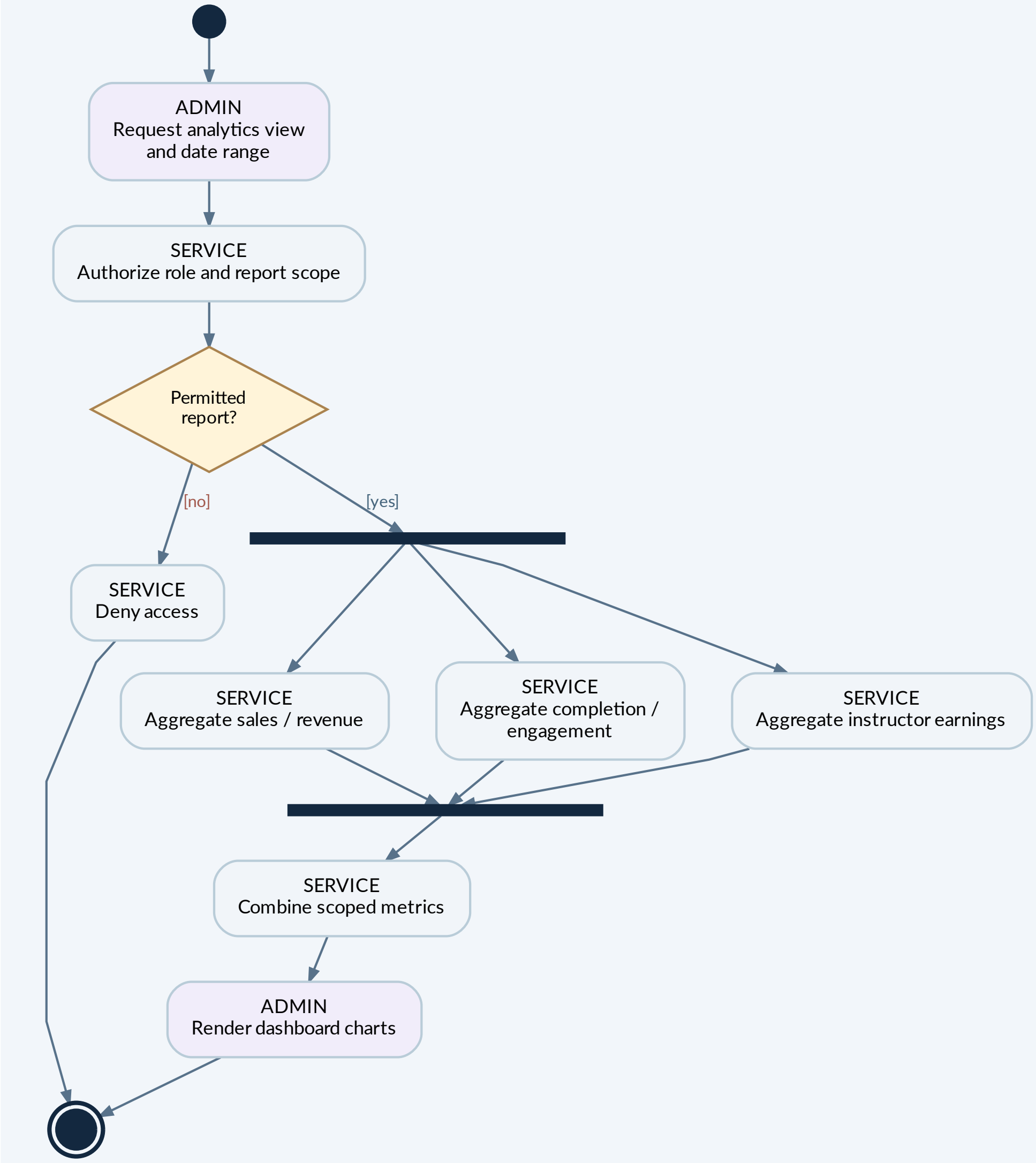
Q&A response or validated rating becomes visible to permitted users.

ALTERNATE / EXCEPTION

Unauthorized or duplicate/ineligible review is rejected.

Build dashboard analytics and reports

AD-14 / REPORTING / UC-34, UC-41 / Brief pp. 3, 5



Admin

Service

NOMINAL OUTCOME

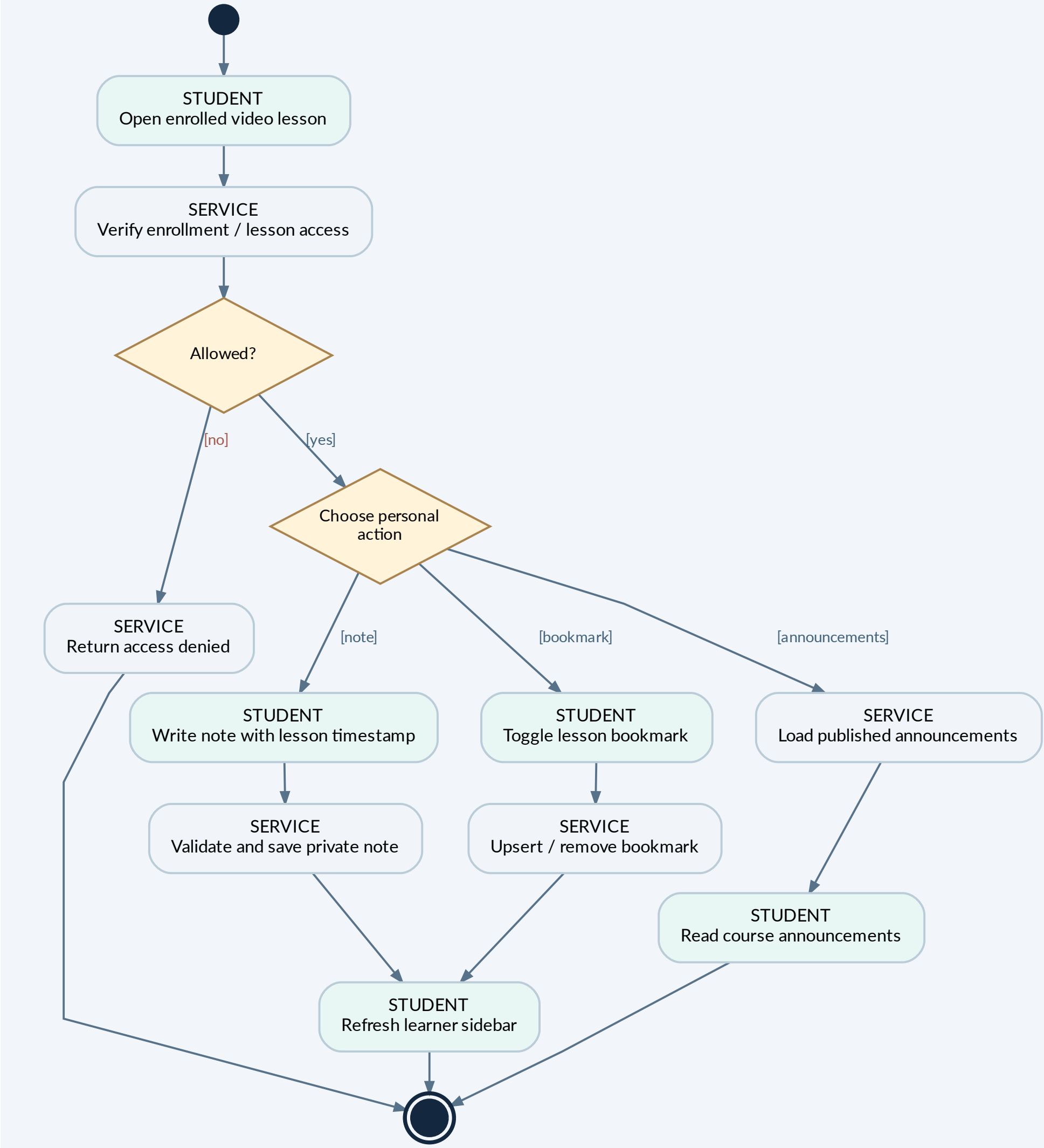
Aggregated revenue, engagement and earnings metrics render as charts.

ALTERNATE / EXCEPTION

Unauthorized scope rejected; absent data produces empty-safe metric output.

Manage lesson notes and bookmarks

AD-15 / PERSONAL LEARNING / UC-17, UC-18, UC-20 / Brief p. 4



Student **Service**

NOMINAL OUTCOME

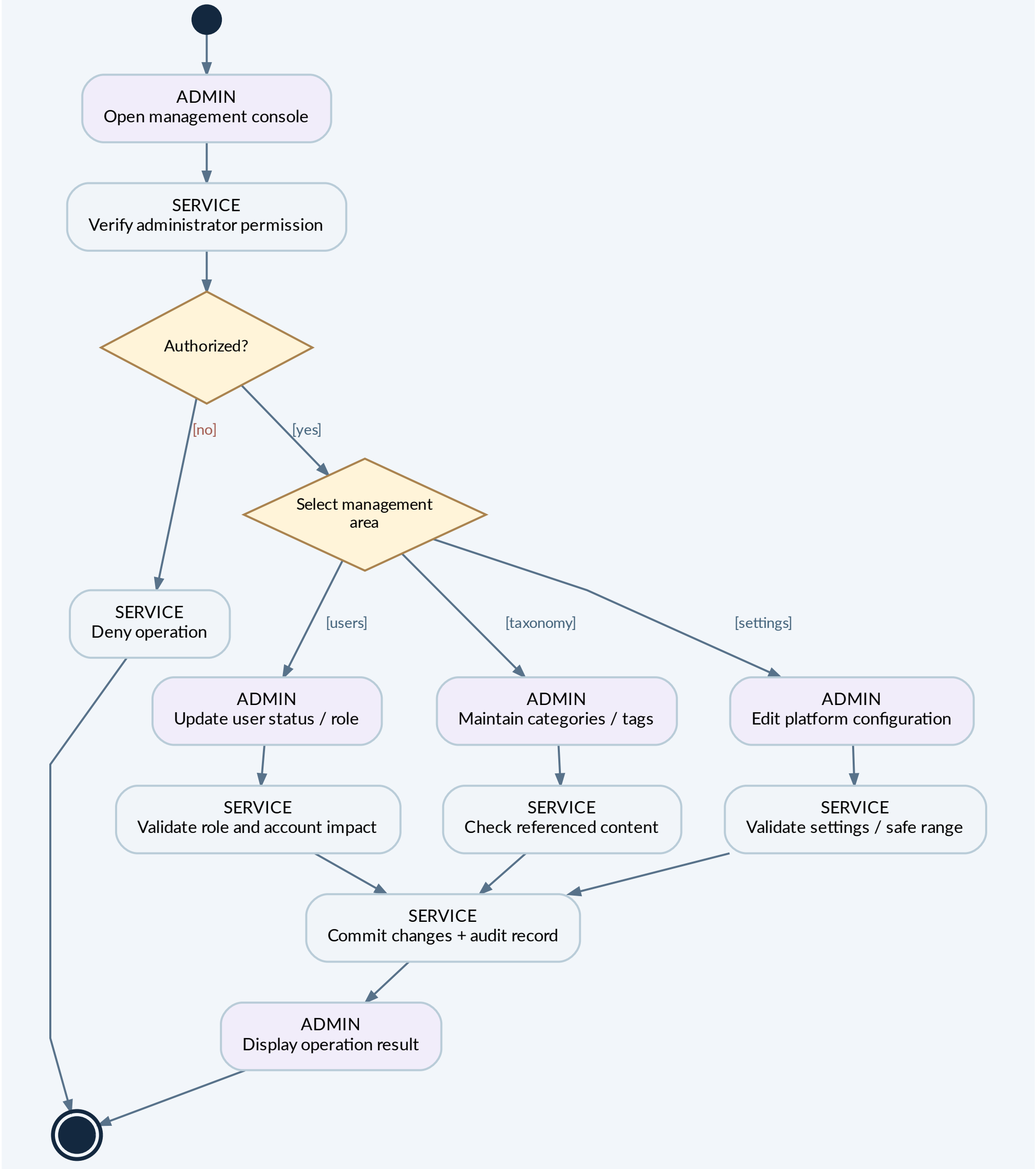
Private note or personal bookmark is updated and visible to its owner.

ALTERNATE / EXCEPTION

Unauthorized actions do not expose or alter learner data.

Admin manage users, taxonomy and settings

AD-16 / PLATFORM GOVERNANCE / UC-39, UC-40, UC-42 / Brief p. 5



Admin

Service

NOMINAL OUTCOME

Validated change is committed with an auditable operation result.

ALTERNATE / EXCEPTION

Insufficient privilege or invalid updates are rejected.

Activity specifications | 01-02

Triggers, preconditions, outcomes, exception handling and data effects accompany each UML view.

AD-01 / Account sign-in & authorization

TRACE UC-02 | FR-001-006

TRIGGER	User attempts to sign in.
SUCCESS	Authenticated user reaches an authorized role-specific entry point.
ALTERNATE / FAILURE	Invalid / locked account shows generic denial; unauthorized resource access is blocked.
GUARD / POLICY	Server validates role and resource access; never trust user-supplied roles.
DATA IMPACT	Identity users, roles, profiles; auth cookie.
DESIGN NOTE	Lockout thresholds and confirmation policy are proposed.

AD-02 / Create or edit a course draft

TRACE UC-25, UC-26, UC-29, UC-30 | FR-014-019

TRIGGER	Instructor creates a course or reopens a draft.
SUCCESS	Course sections, lesson order, pricing and metadata persist as a draft.
ALTERNATE / FAILURE	Ownership violation denies mutation; invalid metadata returns actionable errors.
GUARD / POLICY	Draft save does not imply publication; submission is AD-04.
DATA IMPACT	Courses, CourseSections, Lessons, categories, pricing metadata.
DESIGN NOTE	Drag-and-drop reorder persistence uses proposed stable position values.

Activity specifications | 03-04

Triggers, preconditions, outcomes, exception handling and data effects accompany each UML view.

AD-03 / Upload and process lesson video

TRACE UC-27, UC-28, UC-46 | FR-020-023

TRIGGER	Instructor adds a lesson video.
SUCCESS	Verified, processed video is Ready and addressable via private media metadata.
ALTERNATE / FAILURE	Invalid file, interrupted upload or processor failure leaves no playable Ready asset.
GUARD / POLICY	Only Ready media can be streamed; processing is asynchronous (proposed).
DATA IMPACT	LessonMedia, Azure private blobs, processing jobs/metadata.

DESIGN NOTE The specification requires compression/thumbnails but does not define the processing worker.

AD-04 / Submit, review and publish a course

TRACE UC-31, UC-37, UC-38, UC-48 | FR-024, FR-069-070

TRIGGER	Instructor submits a complete draft.
SUCCESS	Admin-approved course becomes visible in the public catalog.
ALTERNATE / FAILURE	Incomplete submission is blocked; rejected course remains unavailable for sale.
GUARD / POLICY	Only admin approval enables publishing in proposed approval-first policy.
DATA IMPACT	Courses status, approvals, notifications.

DESIGN NOTE Submission/approval state machine reconciles instructor publishing with admin approval.

Activity specifications | 05-06

Triggers, preconditions, outcomes, exception handling and data effects accompany each UML view.

AD-05 / Browse catalog and enroll for free

TRACE UC-03, UC-04, UC-07 | FR-007-012, FR-049

TRIGGER	Learner chooses a catalog course.
SUCCESS	Free course receives one enrollment, without a payment requirement.
ALTERNATE / FAILURE	Paid course routes to checkout; repeat enrollments do not duplicate records.
GUARD / POLICY	Course must be published; free enrollment is idempotent.
DATA IMPACT	Courses, CourseCategories, Enrollments.
DESIGN NOTE	Public catalog access for guest is derived; paid course handling belongs to AD-06.

AD-06 / Cart, discount and Stripe checkout

TRACE UC-08, UC-09, UC-10 | FR-045-053

TRIGGER	Student checks out paid courses.
SUCCESS	Pending order and Stripe checkout session exist; payment is not yet fulfilled.
ALTERNATE / FAILURE	Invalid coupon, stale prices, session error or cancellation does not create paid access.
GUARD / POLICY	Never grant enrollment from client return URL.
DATA IMPACT	Carts, Orders, OrderItems, Coupons, Stripe session reference.
DESIGN NOTE	Repricing and immutable order snapshots are proposed integrity safeguards.

Activity specifications | 07-08

Triggers, preconditions, outcomes, exception handling and data effects accompany each UML view.

AD-07 / Verify Stripe webhook and fulfill order

TRACE UC-43, UC-44, UC-48 | FR-053-055

TRIGGER	Payment provider delivers a signed event.
SUCCESS	Verified payment grants entitlements once and commits atomically.
ALTERNATE / FAILURE	Forged, duplicated, unmatched or failed events do not create duplicate enrollments.
GUARD / POLICY	Persist webhook idempotency key; commit before success acknowledgment.
DATA IMPACT	StripeEvents, Payments, Orders, OrderItems, Enrollments, notifications.
DESIGN NOTE	Exact event types and fulfillment reconciliation policy are team decisions.

AD-08 / Authorize and stream protected lesson

TRACE UC-12, UC-13, UC-47 | FR-026-031

TRIGGER	Enrolled student requests lesson playback.
SUCCESS	Authorized learner receives short-lived access to playable media.
ALTERNATE / FAILURE	Unenrolled user or unready media receives no usable grant.
GUARD / POLICY	Grant is scoped/expiring; reauthorization on refresh.
DATA IMPACT	Enrollments, Lessons, LessonMedia, private blobs / CDN.
DESIGN NOTE	SAS or equivalent token technique is a design choice; do not store public video URLs.

Activity specifications | 09-10

Triggers, preconditions, outcomes, exception handling and data effects accompany each UML view.

AD-09 / Record lesson completion and progress

TRACE UC-14, UC-15, UC-48 | FR-032-036

TRIGGER	Player signals a completion event.
SUCCESS	Progress updates without duplicate counting; completion may trigger certification check.
ALTERNATE / FAILURE	Stale or unauthorized events are rejected and repeated completions are safe.
GUARD / POLICY	Completion is idempotent; quiz gates must also pass before certification.
DATA IMPACT	LessonProgress, Enrollments, course progress aggregates.

DESIGN NOTE Playback evidence thresholds and exact completion formula need product approval.

AD-10 / Take, grade and complete a quiz

TRACE UC-16, UC-33 | FR-037-044

TRIGGER	Student submits answers to an assessment.
SUCCESS	Attempt stored, score displayed, pass status contributes to eligibility.
ALTERNATE / FAILURE	Unauthorized or exhausted attempt blocked; failed quiz offers retake only if policy allows.
GUARD / POLICY	Passing threshold from quiz configuration, not client input.
DATA IMPACT	Quizzes, QuizQuestions, Answers, QuizAttempts, AttemptAnswers.

DESIGN NOTE Attempt limits and retake rules are not specified in the brief.

Activity specifications | 11-12

Triggers, preconditions, outcomes, exception handling and data effects accompany each UML view.

AD-11 / Generate completion certificate

TRACE UC-23, UC-45 | FR-063-066

TRIGGER	Course and quiz completion event is raised.
SUCCESS	One verifiable personalized PDF certificate is registered for the learner.
ALTERNATE / FAILURE	Ineligible, duplicate or failed-storage case issues nothing new.
GUARD / POLICY	Unique certificate code and per enrollment-course uniqueness enforced.
DATA IMPACT	Certificates, Enrollments, CourseProgress, private PDF Blob.
DESIGN NOTE	PDF generator technology is stated; retry / orphan cleanup is proposed.

AD-12 / Public certificate verification and sharing

TRACE UC-05, UC-23, UC-24 | FR-065-068

TRIGGER	Visitor verifies a code or learner shares a certificate.
SUCCESS	Valid credential is publicly verifiable; learner can initiate LinkedIn sharing.
ALTERNATE / FAILURE	Unknown or invalid code returns safe generic feedback.
GUARD / POLICY	Public page must not disclose hidden student data or private storage links.
DATA IMPACT	Certificates and certificate verification index.
DESIGN NOTE	Revocation is a proposed lifecycle policy; not required by the brief.

Activity specifications | 13-14

Triggers, preconditions, outcomes, exception handling and data effects accompany each UML view.

AD-13 / Course Q&A and learner reviews

TRACE UC-19, UC-21, UC-22, UC-35 | FR-056-062

TRIGGER	Student posts a question or course review.
SUCCESS	Q&A response or validated rating becomes visible to permitted users.
ALTERNATE / FAILURE	Unauthorized or duplicate/ineligible review is rejected.
GUARD / POLICY	Review range 1..5; sanitize learner-contributed text.
DATA IMPACT	CourseQuestions, CourseAnswers, CourseReviews, notifications.

DESIGN NOTE Review uniqueness and instructor response service-level controls are proposed.

AD-14 / Build dashboard analytics and reports

TRACE UC-34, UC-41 | FR-074-083

TRIGGER	Admin or scoped instructor selects an analytics dashboard.
SUCCESS	Aggregated revenue, engagement and earnings metrics render as charts.
ALTERNATE / FAILURE	Unauthorized scope rejected; absent data produces empty-safe metric output.
GUARD / POLICY	A fork/join depicts logical parallel aggregations, not a required multithreaded implementation.
DATA IMPACT	Orders, Payments, Enrollments, Progress, instructor earnings.

DESIGN NOTE Cache policy and report freshness SLA are proposed.

Activity specifications | 15-16

Triggers, preconditions, outcomes, exception handling and data effects accompany each UML view.

AD-15 / Manage lesson notes and bookmarks

TRACE UC-17, UC-18, UC-20 | FR-032-036 (related)

TRIGGER	Student saves a lesson note or bookmark.
SUCCESS	Private note or personal bookmark is updated and visible to its owner.
ALTERNATE / FAILURE	Unauthorized actions do not expose or alter learner data.
GUARD / POLICY	Notes are private by default and scoped to enrollment owner.
DATA IMPACT	LessonNotes, LessonBookmarks, Enrollments.
DESIGN NOTE	Timestamp unit, edit policy and note limits are proposed.

AD-16 / Admin manage users, taxonomy and settings

TRACE UC-39, UC-40, UC-42 | FR-069-083

TRIGGER	Administrator modifies users, course taxonomy or platform settings.
SUCCESS	Validated change is committed with an auditable operation result.
ALTERNATE / FAILURE	Insufficient privilege or invalid updates are rejected.
GUARD / POLICY	Sensitive role changes should be auditable and protected from self-escalation.
DATA IMPACT	Identity users/roles, Categories, Tags, PlatformSettings, audit.
DESIGN NOTE	Audit details and settings schema are proposed implementation rules.

Cross-cutting controls & token safety

Behavioral invariants applied throughout the proposed EduHub workflow model.

01 AUTHORIZATION

An authenticated cookie alone is not course entitlement. Check course ownership, role, enrollment and lesson membership server-side.

02 PAYMENT TRUST

The browser return page does not prove payment. Signed Stripe events trigger order reconciliation and idempotent enrollment only after commit.

03 MEDIA IS PRIVATE

Azure uploads and reads use narrowly scoped, short-lived permissions. A Pending or Failed media record is never treated as Ready.

04 PROGRESS IS IDEMPOTENT

Repeated completion signals do not inflate the number of completed lessons or issue duplicate course-completion events.

05 CERTIFICATE ELIGIBILITY

The server checks full curriculum and quiz policy before generating one unique, verifiable certificate for an eligible learner.

06 ADMIN SCOPE

Published content and sensitive role/configuration changes are controlled, recorded and excluded from self-service elevation.

Traceability | behavior to use cases

Stable cross-document identifiers keep the activity model aligned with the previous requirements and interaction set.

AD-01	Account sign-in & authorization	UC-02	Brief pp. 1-2
AD-02	Create or edit a course draft	UC-25, UC-26, UC-29, UC-30	Brief pp. 1, 3
AD-03	Upload and process lesson video	UC-27, UC-28, UC-46	Brief p. 3
AD-04	Submit, review and publish a course	UC-31, UC-37, UC-38, UC-48	Brief pp. 3, 5
AD-05	Browse catalog and enroll for free	UC-03, UC-04, UC-07	Brief pp. 1, 4
AD-06	Cart, discount and Stripe checkout	UC-08, UC-09, UC-10	Brief p. 4
AD-07	Verify Stripe webhook and fulfill order	UC-43, UC-44, UC-48	Brief p. 4
AD-08	Authorize and stream protected lesson	UC-12, UC-13, UC-47	Brief pp. 3-4
AD-09	Record lesson completion and progress	UC-14, UC-15, UC-48	Brief p. 4
AD-10	Take, grade and complete a quiz	UC-16, UC-33	Brief p. 5
AD-11	Generate completion certificate	UC-23, UC-45	Brief p. 5
AD-12	Public certificate verification and sharing	UC-05, UC-23, UC-24	Brief p. 5
AD-13	Course Q&A and learner reviews	UC-19, UC-21, UC-22, UC-35	Brief pp. 3, 5
AD-14	Build dashboard analytics and reports	UC-34, UC-41	Brief pp. 3, 5
AD-15	Manage lesson notes and bookmarks	UC-17, UC-18, UC-20	Brief p. 4
AD-16	Admin manage users, taxonomy and settings	UC-39, UC-40, UC-42	Brief p. 5

Review & validation checklist

Apply these checks when finalizing the baseline with the supervisor and implementation team.

- [ ] 01 Every branch guard is mutually exclusive or has an explicit fallback.
- [ ] 02 Each action has a responsible actor or service and reaches a terminal outcome.
- [ ] 03 Webhooks: no paid entitlement until verified payment is committed.
- [ ] 04 Retransmission: duplicate payment events and progress signals are harmless.
- [ ] 05 Content: course approval and media Ready status gate public availability.
- [ ] 06 Authorization: student enrollment, instructor ownership, admin roles checked.
- [ ] 07 Certificate: only one verified issue per eligible student/course combination.
- [ ] 08 Retries: failed media/PDF generation can be resumed without double effects.
- [ ] 09 Fork/join in AD-14 depicts logical independence, not a required implementation.
- [ ] 10 Changes to approval, retake, coupon and revocation policy require approval.

Sources, standards & modeling notes

Primary project evidence is distinct from third-party technical guidance.

S1 **EduHub - Online Learning Platform, supplied graduation project brief, 5 pages, pp. 1-5.**

User-supplied project PDF.

S2 **EduHub - Use Case Diagram & Descriptions; 48 use cases and 14 full scenarios.**

Earlier companion design deliverable.

S3 **EduHub - Functional & Non-Functional Requirements; 83 FR and 32 NFR.**

Earlier requirements baseline.

S4 **EduHub - Sequence Diagrams; 14 interaction views and traceability matrix.**

Earlier interaction companion document.

R1 **Object Management Group, UML Specification 2.5.1.**

https://www.omg.org/spec/UML/2.5.1/

R2 **Graphviz - DOT language reference (editable diagram sources).**

https://graphviz.org/doc/info/lang.html

R3 **Stripe, Checkout fulfillment guidance.**

https://docs.stripe.com/checkout/fulfillment

R4 **Microsoft Learn, Azure Storage SAS overview.**

https://learn.microsoft.com/en-us/azure/storage/common/storage-sas-overview